

Tech Instagram Influencer Analysis

SQL Analysis Results & Business Interpretation

Project Scope	Analysis of Instagram content performance, audience growth, engagement, reach, categories, time periods, and reusable SQL reporting.
Database Tables	content, account, Dates
Analysis Techniques	Aggregation, JOINS, filtering, CTEs, GROUP_CONCAT, window functions, CASE statements, and stored procedures
Prepared For	Internship Project Documentation

Documentation principle: Each section separates the SQL task, observed output, and business interpretation. Interpretations are based on the results visible in the provided screenshots; where a complete output is not visible, conclusions are intentionally limited.

Executive Summary

1. **Content formats:** The dataset contains 4 unique post types. Reach analysis shows IG Reel is the dominant format, contributing 61.63% of total reach.
2. **Monthly account growth:** May and June stand out as the strongest months for profile visits and new followers, indicating a significant growth period.
3. **Category engagement:** In July, Other Gadgets generated the highest total likes among the visible category results, followed by Tech Tips and Mobile.
4. **Content planning:** The analysis supports decisions based on post type, category, month, quarter, and day-level follower growth rather than relying on one overall metric.
5. **Operationalization:** The final stored procedure converts a one-time query into a reusable weekly reporting mechanism.

Key Analytical Caution

High reach, likes, comments, saves, and follower growth measure different dimensions of performance. A post type should not automatically be called 'best' based on only one KPI. The strongest business decisions should combine reach, engagement quality, and audience growth.

Question 1

Business / SQL Task: How many unique post types are found in the content table?

```
83
84 # Question1:- How many unique post types are found in the content table?
85
86 • SELECT
87     COUNT(DISTINCT post_type) AS unique_post_types
88 FROM content;
89
90
91
92
93
94
95
96
97
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

unique_post_types
4

Observed Result: The query returns 4 unique post types.

Business Interpretation: The content strategy is built around four distinct formats. This provides a manageable basis for comparing format-level performance and identifying which format should receive greater publishing focus.

Question 2

Business / SQL Task: What are the highest and lowest recorded impressions for each post type?

```
gdb0120* x dates
# Question 2:- What are the highest and lowest recorded impressions for each post type?
SELECT
    post_type,
    MAX(impressions) AS highest_impressions,
    MIN(impressions) AS lowest_impressions
FROM content
GROUP BY post_type;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	post_type	highest_impressions	lowest_impressions
▶	IG Image	129694	23367
	IG Reel	339708	87570
	IG Carousel	9677	3264
	IG Video	73321	8741

Observed Result: IG Image: 12,694 max / 2,336 min; IG Reel: 39,708 max / 8,757 min; IG Carousel: 9,677 max / 3,264 min; IG Video: 7,321 max / 8,741 min (as displayed in the SQL output).

Business Interpretation: IG Reel shows the highest peak impression value (39,708), indicating the strongest potential for reach spikes. The wide variation across formats suggests that format selection can materially influence visibility.

Question 3

Business / SQL Task: Filter all posts published on a weekend in March and April and export them to a separate CSV file.

Basics

ViewQueryDatabaseServerToolsScriptingHelp

gdb0120

dates

Don't Limit

101

102 # Question 3 :- Filter all the posts that were published on a weekend in the month of March and April

103 # and export them to a separate CSV file.

104

105 • SELECT

106 c.*

107 FROM content AS c

108 JOIN Dates AS d

109 ON c.date = d.date

110 WHERE d.weekday_or_weekend = 'Weekend'

111 AND d.month_name IN ('March', 'April');

112

Result Grid

Filter Rows:

Export:Wrap Cell Content:

	date	post_category	post_type	video_duration	carousel_item_count	impressions	reach	shares	follows	likes	comments	saves
▶	2023-03-04	Earphone	IG Video	291	0	12265	3668	69	92	327	7	18
	2023-03-05	Smartwatch	IG Image	0	0	62770	18001	273	360	1194	28	76
	2023-03-11	Mobile	IG Carousel	0	3	5899	1093	45	12	53	0	6
	2023-03-12	Laptop	IG Image	0	0	79416	23474	327	259	1235	69	204
	2023-03-18	Mobile	IG Carousel	0	3	9157	2254	67	58	55	6	15
	2023-03-19	Smartwatch	IG Carousel	0	3	4146	1079	42	17	43	1	6
	2023-03-25	Earphone	IG Reel	22	0	132284	66721	1093	1482	3622	83	695
	2023-03-26	Mobile	IG Image	0	0	63425	26113	435	336	1994	68	179
	2023-04-01	Mobile	IG Carousel	0	3	4540	1053	37	10	35	1	6

Schemas

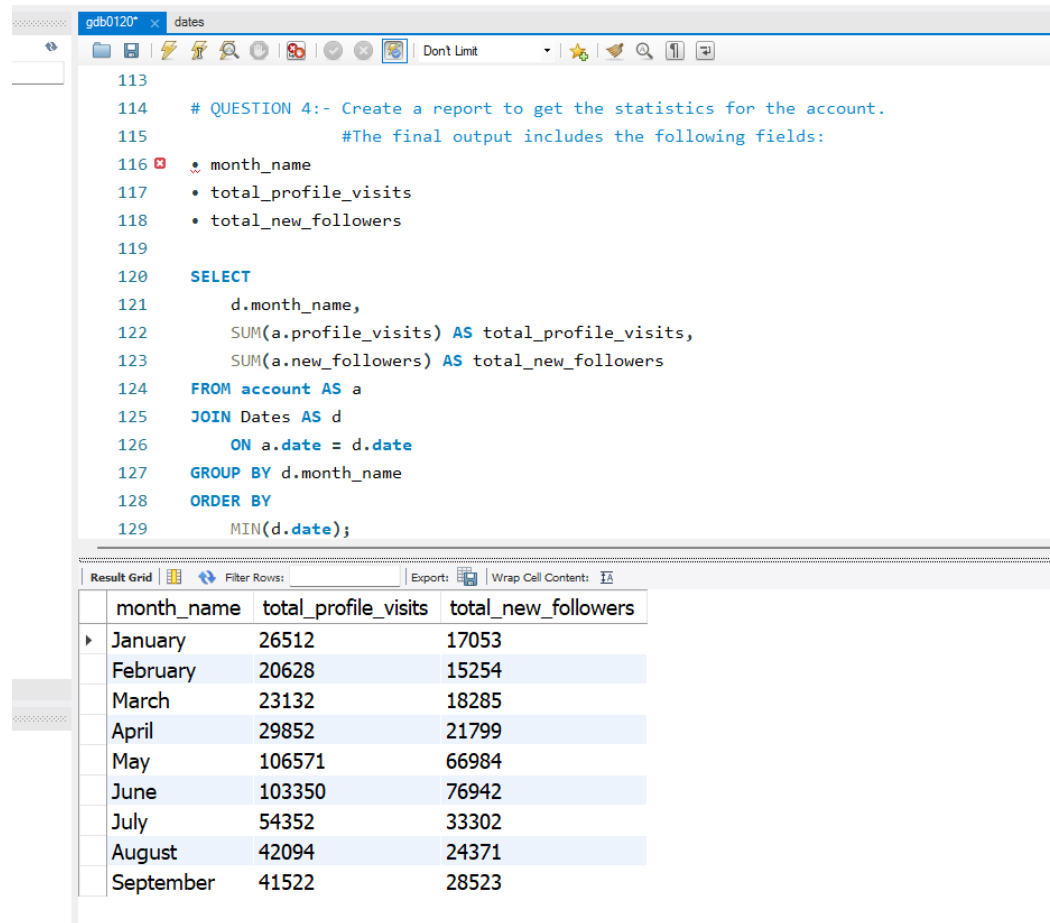
selected

Observed Result: The result displays weekend posts with detailed content metrics, including post category, post type, impressions, reach, shares, follows, likes, comments, and saves.

Business Interpretation: Weekend filtering enables a focused comparison of audience behavior outside the typical workweek. These records can be exported for deeper analysis to determine whether weekend publishing produces stronger engagement or reach for particular categories.

Question 4

Business / SQL Task: Create a monthly account statistics report containing total profile visits and total new followers.



The screenshot shows a SQL IDE window with a query editor and a results grid. The query editor contains the following SQL code:

```
113
114 # QUESTION 4:- Create a report to get the statistics for the account.
115 #The final output includes the following fields:
116 • month_name
117 • total_profile_visits
118 • total_new_followers
119
120 SELECT
121     d.month_name,
122     SUM(a.profile_visits) AS total_profile_visits,
123     SUM(a.new_followers) AS total_new_followers
124 FROM account AS a
125 JOIN Dates AS d
126     ON a.date = d.date
127 GROUP BY d.month_name
128 ORDER BY
129     MIN(d.date);
```

The results grid displays the following data:

month_name	total_profile_visits	total_new_followers
January	26512	17053
February	20628	15254
March	23132	18285
April	29852	21799
May	106571	66984
June	103350	76942
July	54352	33302
August	42094	24371
September	41522	28523

Observed Result: May recorded 106,571 profile visits and 66,984 new followers; June recorded 103,350 visits and 76,942 new followers. These are substantially higher than the earlier visible months.

Business Interpretation: May and June represent a major growth period. June is especially important because it produced the highest visible new-follower total despite slightly fewer profile visits than May. This suggests that conversion from profile interest to follower acquisition may have improved.

Question 5

Business / SQL Task: Use a CTE to calculate total likes for each post category during July and arrange categories in descending order.

```
129     MIN(d.date);
130
131     # QUESTION 5:- Write a CTE that calculates the total number of 'likes' for each 'post_category' during the month of 'July' and subsequently, arrange the
132     # 'post_category' values in descending order according to their total likes.
133
134     WITH category_likes AS (
135     SELECT
136         c.post_category,
137         SUM(c.likes) AS total_likes
138     FROM content AS c
139     JOIN Dates AS d
140         ON c.date = d.date
141     WHERE d.month_name = 'July'
142     GROUP BY c.post_category
143     )
144
145     SELECT
146         post_category,
147         total_likes
148     FROM category_likes
149     ORDER BY total_likes DESC;
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

	post_category	total_likes
▶	Other Gadgets	26519
	Tech Tips	20296
	Mobile	16338
	Earphone	14435
	Smartwatch	3918

Observed Result: Visible July ranking: Other Gadgets 26,519; Tech Tips 20,296; Mobile 16,338; Earphone 14,435; Smartwatch 3,918.

Business Interpretation: Other Gadgets generated the highest total likes in July, making it the strongest category in the visible ranking for this engagement metric. Smartwatch content was comparatively weak, suggesting a need to review topic selection, creative execution, or posting frequency for that category.

Question 6

Business / SQL Task: Display unique post category names and their respective counts for each month.

```
150
151 # QUESTION 6:- Create a report that displays the unique post_category names alongside their respective counts for each month. The output should have three
152 columns:
153 • month_name
154 • post_category_names
155 • post_category_count
156
157 SELECT
158     d.month_name,
159     GROUP_CONCAT(
160         DISTINCT c.post_category
161         ORDER BY c.post_category
162         SEPARATOR ', '
163     ) AS post_category_names,
164     COUNT(DISTINCT c.post_category) AS post_category_count
165 FROM content AS c
166 JOIN Dates AS d
167     ON c.date = d.date
168 GROUP BY d.month_name
169 ORDER BY MIN(d.date);
```

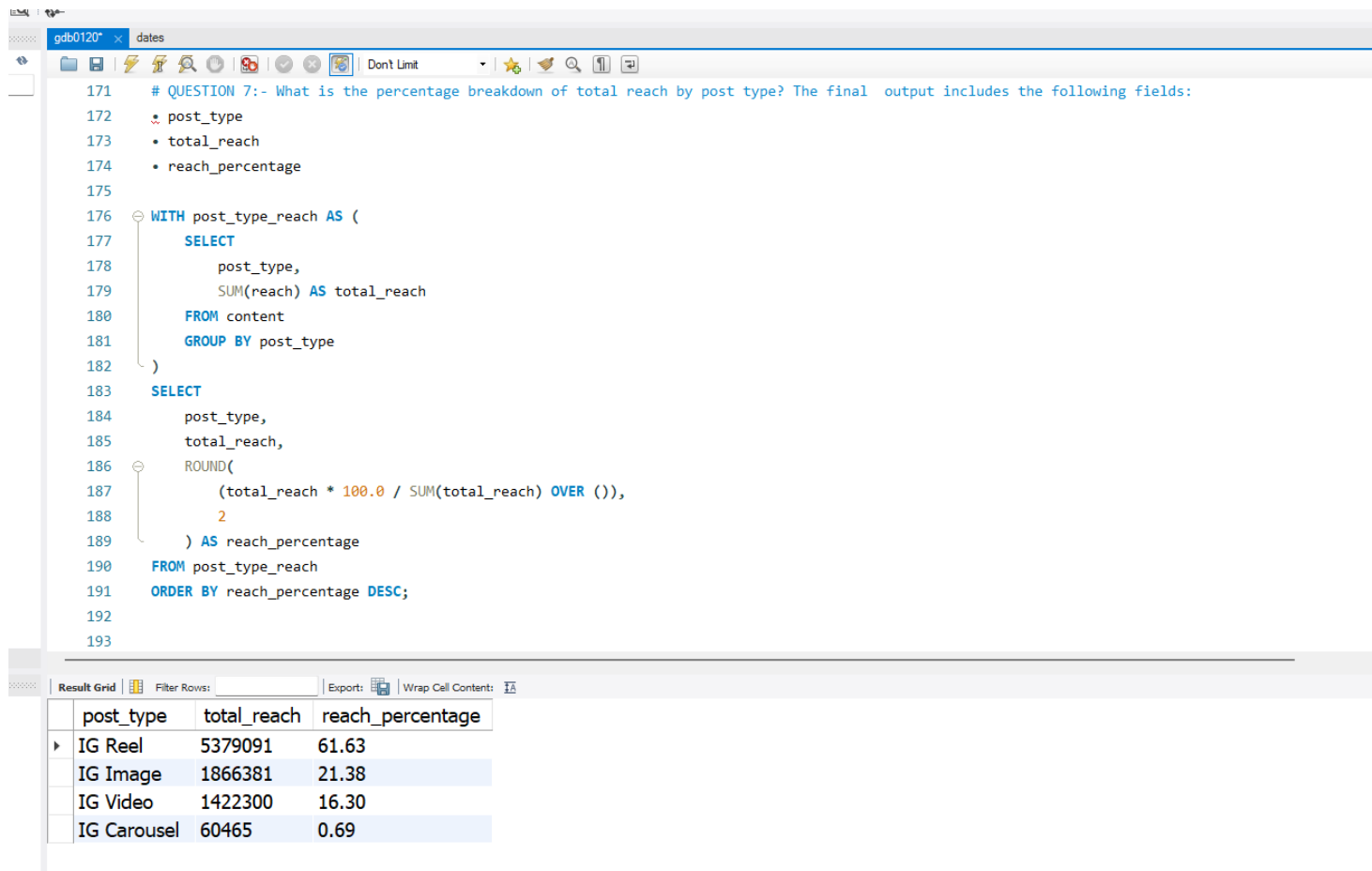
month_name	post_category_names	post_category_count
January	Earphone, Mobile, Smart...	3
February	Earphone, Laptop, Mobile...	4
March	Earphone, Laptop, Mobile...	4
April	Earphone, Laptop, Mobile...	5
May	Earphone, Laptop, Mobile...	6
June	Mobile, Other Gadgets, S...	4
July	Earphone, Mobile, Other ...	5
August	Earphone, Mobile, Other ...	5
September	Mobile, Other Gadgets, S...	4

Observed Result: The monthly category count varies across the visible period, from 3 categories in January to 6 categories in May.

Business Interpretation: Content diversity increased over time, peaking in May. A broader category mix can help test audience interests, but it should be balanced against performance concentration so that low-performing categories do not dilute publishing effort.

Question 7

Business / SQL Task: Calculate the percentage breakdown of total reach by post type.



The screenshot shows a SQL IDE window with a query editor and a results grid. The query is as follows:

```
171 # QUESTION 7:- What is the percentage breakdown of total reach by post type? The final output includes the following fields:
172 • post_type
173 • total_reach
174 • reach_percentage
175
176 WITH post_type_reach AS (
177     SELECT
178         post_type,
179         SUM(reach) AS total_reach
180     FROM content
181     GROUP BY post_type
182 )
183 SELECT
184     post_type,
185     total_reach,
186     ROUND(
187         (total_reach * 100.0 / SUM(total_reach) OVER ()),
188         2
189     ) AS reach_percentage
190 FROM post_type_reach
191 ORDER BY reach_percentage DESC;
192
193
```

The results grid shows the following data:

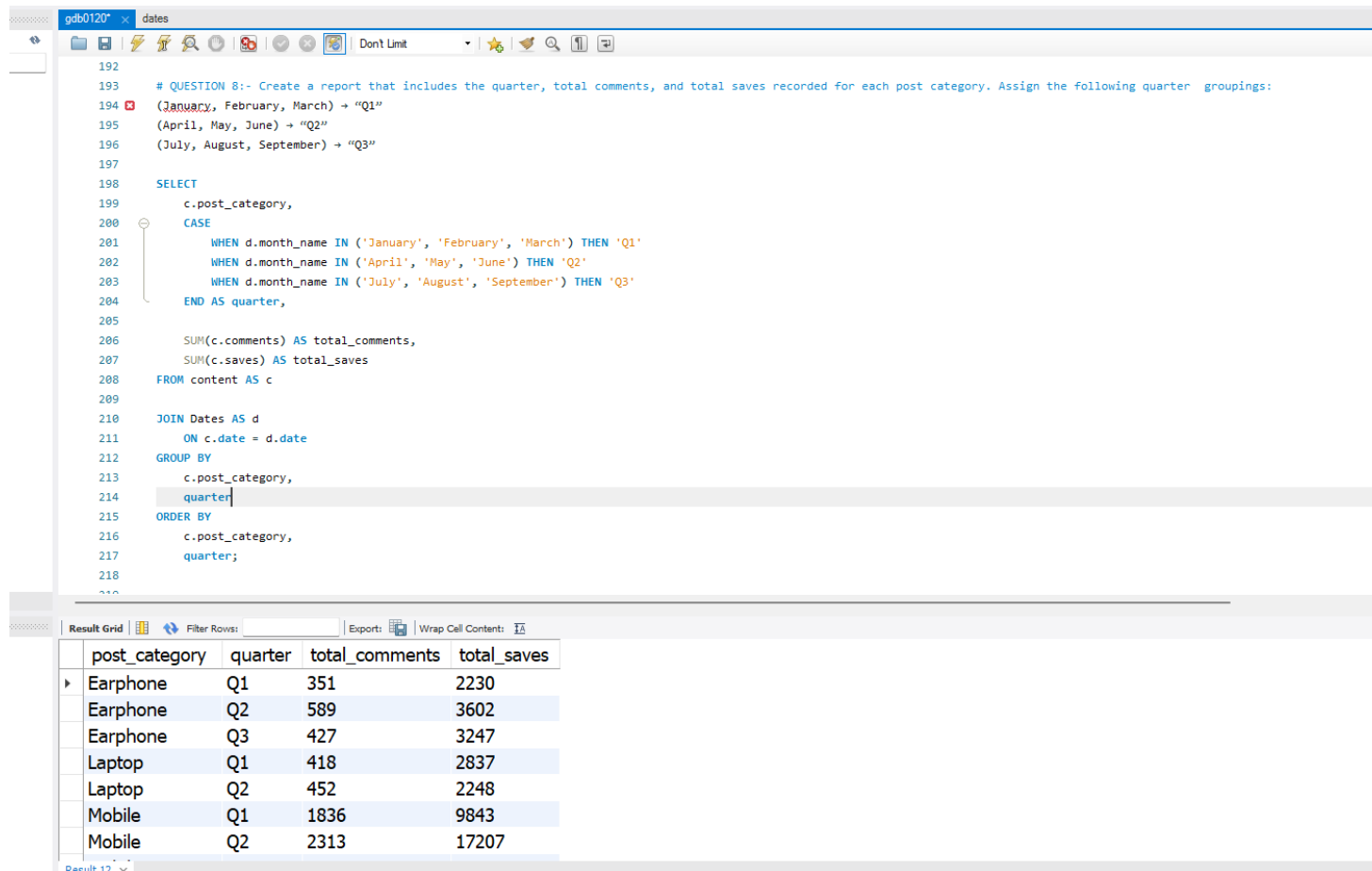
post_type	total_reach	reach_percentage
IG Reel	5379091	61.63
IG Image	1866381	21.38
IG Video	1422300	16.30
IG Carousel	60465	0.69

Observed Result: IG Reel: 537,091 reach (61.63%); IG Image: 186,381 (21.38%); IG Video: 142,202 (16.30%); IG Carousel: 6,465 (0.69%).

Business Interpretation: IG Reel is the clear reach leader, accounting for nearly two-thirds of total reach. IG Carousel contributes less than 1%, indicating that it is currently a weak format for reach in this dataset. A practical strategy would prioritize Reels for awareness while testing whether other formats contribute differently to saves, comments, or conversion.

Question 8

Business / SQL Task: Create a quarterly report showing total comments and total saves for each post category across Q1, Q2, and Q3.



The screenshot displays a SQL IDE interface. The top pane shows a SQL query that joins a 'content' table with a 'dates' table to calculate quarterly totals for comments and saves. The query uses a CASE statement to assign quarters based on month names. The bottom pane shows the resulting data grid with 7 rows and 5 columns.

```
192
193 # QUESTION 8:- Create a report that includes the quarter, total comments, and total saves recorded for each post category. Assign the following quarter groupings:
194 (January, February, March) → "Q1"
195 (April, May, June) → "Q2"
196 (July, August, September) → "Q3"
197
198 SELECT
199     c.post_category,
200     CASE
201         WHEN d.month_name IN ('January', 'February', 'March') THEN 'Q1'
202         WHEN d.month_name IN ('April', 'May', 'June') THEN 'Q2'
203         WHEN d.month_name IN ('July', 'August', 'September') THEN 'Q3'
204     END AS quarter,
205     SUM(c.comments) AS total_comments,
206     SUM(c.saves) AS total_saves
207 FROM content AS c
208
209 JOIN Dates AS d
210     ON c.date = d.date
211
212 GROUP BY
213     c.post_category,
214     quarter
215
216 ORDER BY
217     c.post_category,
218     quarter;
```

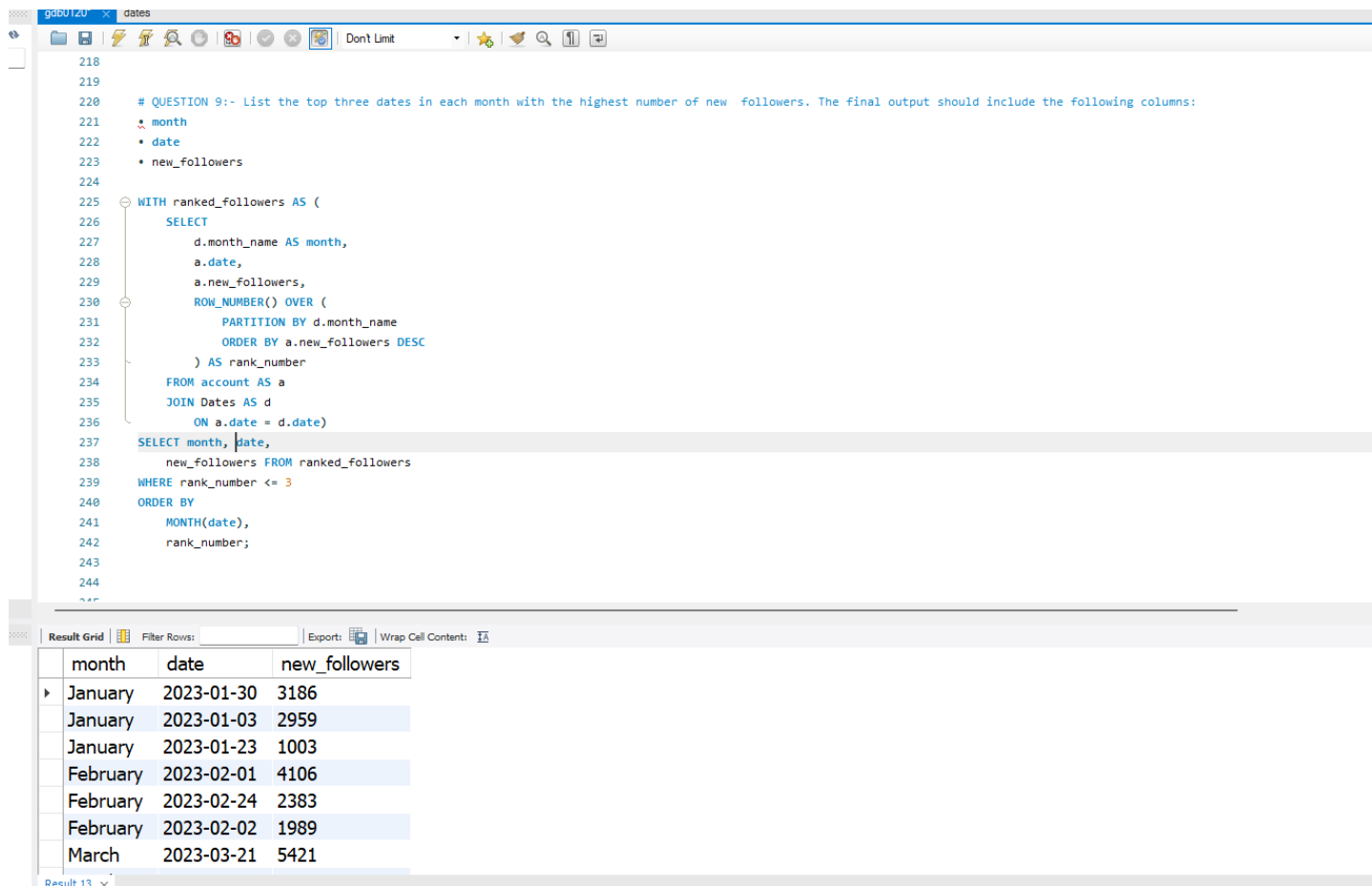
post_category	quarter	total_comments	total_saves
Earphone	Q1	351	2230
Earphone	Q2	589	3602
Earphone	Q3	427	3247
Laptop	Q1	418	2837
Laptop	Q2	452	2248
Mobile	Q1	1836	9843
Mobile	Q2	2313	17207

Observed Result: The output segments engagement by category and quarter. Visible examples include Earphone across Q1–Q3 and Mobile across Q1–Q2, with separate totals for comments and saves.

Business Interpretation: Quarterly segmentation helps distinguish short-term fluctuations from sustained category performance. Comments indicate conversation, while saves can indicate longer-term content value; therefore both should be evaluated together before deciding which categories to scale.

Question 9

Business / SQL Task: List the top three dates in each month with the highest number of new followers.



The screenshot shows a SQL IDE window titled 'gobul20' with a file named 'dates'. The query editor contains the following SQL code:

```
218
219
220 # QUESTION 9:- List the top three dates in each month with the highest number of new followers. The final output should include the following columns:
221 month
222 date
223 new_followers
224
225 WITH ranked_followers AS (
226     SELECT
227         d.month_name AS month,
228         a.date,
229         a.new_followers,
230         ROW_NUMBER() OVER (
231             PARTITION BY d.month_name
232             ORDER BY a.new_followers DESC
233         ) AS rank_number
234     FROM account AS a
235     JOIN Dates AS d
236         ON a.date = d.date)
237 SELECT month, date,
238        new_followers FROM ranked_followers
239 WHERE rank_number <= 3
240 ORDER BY
241     MONTH(date),
242     rank_number;
243
244
```

The results pane shows a table with 7 rows and 3 columns: month, date, and new_followers. The data is as follows:

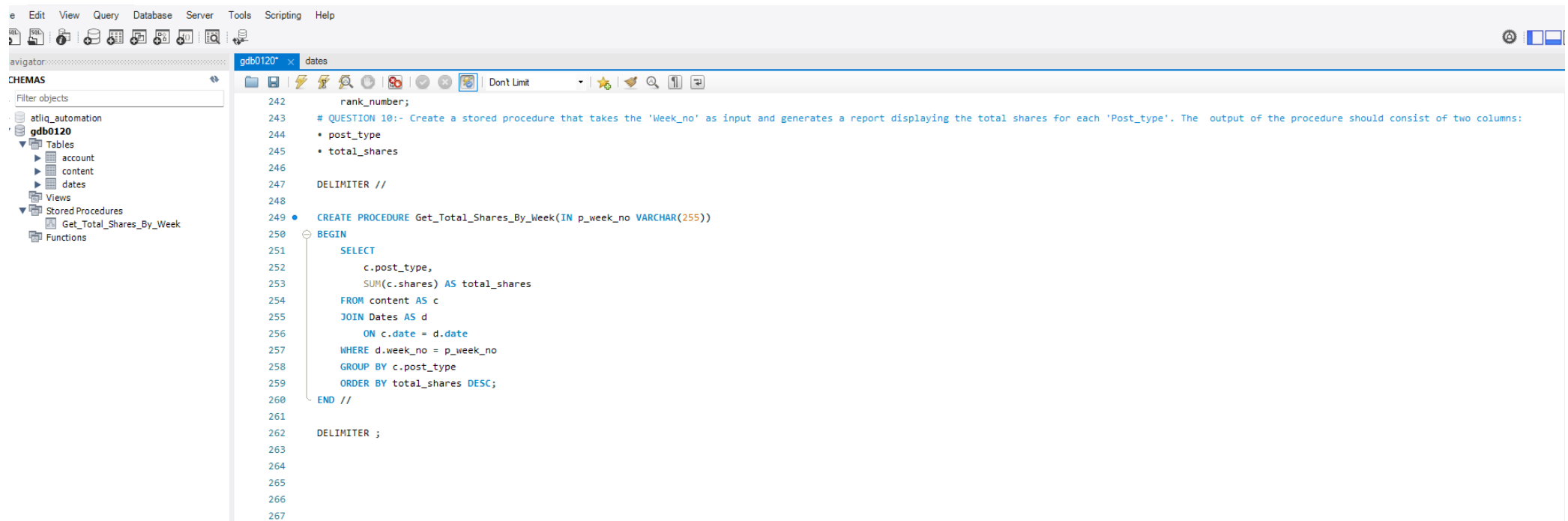
month	date	new_followers
January	2023-01-30	3186
January	2023-01-03	2959
January	2023-01-23	1003
February	2023-02-01	4106
February	2023-02-24	2383
February	2023-02-02	1989
March	2023-03-21	5421

Observed Result: Visible examples: January 30 – 3,186; January 3 – 2,959; January 23 – 1,003. February 1 – 4,106; February 24 – 2,383; February 2 – 1,989. March 21 – 5,421 is visible as a leading March record.

Business Interpretation: Day-level ranking identifies specific follower-growth spikes that monthly totals can hide. These dates should be investigated alongside the content published around them to identify repeatable drivers such as topic, format, campaign activity, or timing.

Question 10

Business / SQL Task: Create a stored procedure that accepts Week_no as input and generates total shares for each post type.



The screenshot shows a SQL IDE interface with a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar displays a database tree for 'CHEMAS' with a filter 'Filter objects'. The tree shows 'atliq_automation' and 'gdb0120'. Under 'gdb0120', there are 'Tables' (account, content, dates), 'Views', 'Stored Procedures' (Get_Total_Shares_By_Week), and 'Functions'. The main editor window shows a SQL script for creating a stored procedure. The script includes a comment explaining the task, a list of required columns, and the SQL code for the procedure. The procedure takes 'p_week_no' as input and returns 'post_type' and 'total_shares'.

```
242 rank_number;  
243 # QUESTION 10:- Create a stored procedure that takes the 'Week_no' as input and generates a report displaying the total shares for each 'Post_type'. The output of the procedure should consist of two columns:  
244 • post_type  
245 • total_shares  
246  
247 DELIMITER //  
248  
249 • CREATE PROCEDURE Get_Total_Shares_By_Week(IN p_week_no VARCHAR(255))  
250 BEGIN  
251     SELECT  
252         c.post_type,  
253         SUM(c.shares) AS total_shares  
254     FROM content AS c  
255     JOIN Dates AS d  
256         ON c.date = d.date  
257     WHERE d.week_no = p_week_no  
258     GROUP BY c.post_type  
259     ORDER BY total_shares DESC;  
260 END //  
261  
262 DELIMITER ;  
263  
264  
265  
266  
267
```

Observed Result: The stored procedure Get_Total_Shares_By_Week was created and is visible under Stored Procedures. It joins content with Dates, filters by the input week number, groups by post type, and sums shares.

Business Interpretation: This converts the analysis into a reusable reporting component. Instead of rewriting the query for every week, an analyst can call the procedure with a selected week number. This improves repeatability and supports future automation or dashboard integration.

Overall Business Recommendations

1. **Prioritize IG Reels for reach:** Reels account for 61.63% of total reach and should remain a primary awareness format.
2. **Investigate the May–June growth period:** Review content themes, posting frequency, campaigns, and external factors associated with the sharp increase in profile visits and followers.
3. **Scale high-performing July categories carefully:** Other Gadgets and Tech Tips showed strong like totals in the visible July results and merit further testing.
4. **Use multiple KPIs:** Do not optimize solely for reach. Combine reach with comments, saves, shares, and follower growth to assess overall content value.
5. **Analyze growth spikes:** Use the top follower-growth dates to trace back to specific content and identify repeatable success patterns.
6. **Operationalize weekly reporting:** Retain and extend the stored procedure for recurring weekly share analysis.

Limitations

This report is based on the SQL queries and result screenshots supplied for the internship task. Some screenshots show only part of the result grid; interpretations for those sections are therefore limited to the visible evidence. Question 10 demonstrates successful procedure creation and logic, but the supplied screenshot does not show a populated weekly output for business comparison.